

Form-stable paraffin/graphene aerogel/copper foam composite phase change material for solar energy conversion and storage

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ABSTRACT

Sustainable utilization of solar energy with a more compact solar heating system is a significant research. In this study, a novel form-stable composite phase change material (PCM) was prepared for application in solar energy conversion and storage. Paraffin as PCM was encapsulated in the matrix consist of copper foam (CF) loaded with graphene aerogel (GA) in order to improve thermal conductivity, prevent leakage and convert solar energy to thermal energy. The melting temperature and the latent heat of the composite PCMs were determined to estimate thermal storage performance. The thermal conductivities of composite PCMs measured by transient plane heat source method were more than 9 times higher than that of paraffin. Besides, the composite PCMs showed uniform temperature distribution and perfect shape-stabilization when heated. Furthermore, the composite PCMs exhibited an excellent light-to-thermal energy conversion efficiency (up to 97%) in the experiment of light-to-thermal energy conversion, which demonstrated the potential of the composite PCM for solar energy conversion and storage.

1. Introduction

Solar energy having characteristics of widespread distribution, cost-free and cleanness to environment is recognized as an excellent clean and renewable source of energy [1,2]. However, extensive utilization of solar energy is embedded by the time- and weather-dependent nature of solar radiation [3]. The inherent intermittent working mode of many solar heating systems will create a large gap between energy supply and demand. Latent thermal energy storage (LTES) is a promising technology to bridge this gap [4,5]. With assistance of phase change material (PCM), LTES system is capable of storing a large quantity of energy as the form of latent heat when solar energy is overloaded and retrieving the heat when solar radiation suspends, and thus a sustainable utilization of solar energy can be ensured [6]. In most cases, absorber is needed to capture solar radiation and convert solar energy to thermal energy. The energy storage efficiency of LTES is limited by how effectively the heat is transferred from absorber to LTES system [7]. Therefore, absorbing the solar energy within the LTES system directly is capable of

not only simplifying the solar heating system but also maximizing the utilization efficiency of solar energy.

In a LTES system, PCM is the core. PCMs integrated in the LTES systems should meet the criteria of adjustable phase change temperature and high latent heat of fusion to ensure suitable working temperature and large heat storage capacity for the system [8]. Paraffin is an attractive kind of PCMs, which has advantages of good thermal and chemical stability and negligible super cooling degree [9,10]. As a mixture of alkanes with different chain lengths and small amounts of other hydrocarbon compounds [11], the phase change temperature and latent heat of paraffin can be adjusted to satisfy different application requirements by changing the chain lengths of alkanes and the addition of hydrocarbon compounds [12]. However, the thermal conductivity of paraffin is about 0.2–0.4 W/(m·K), which is far from sufficiency to ensure high rate of heat storing and releasing for LTES systems [13]. Besides, paraffin will come across leakage problem during its solid-liquid phase transition process, resulting in more complex structures of LTES system [14]. Moreover, paraffin does not have the function

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of light-to-thermal energy conversion. How to improve the thermal conductivity and prevent the leakage of liquid paraffin, as well as convert the light energy to thermal energy in the paraffin still remain great challenges for researches.

There have been intensive efforts in exploring methods to improve the thermal conductivity of paraffin and encapsulate it. Microencapsulation technology is one of the approaches to solve above problems effectively [15]. Generally, paraffin is used as the core material and microencapsulated in the shell commonly made of polymer materials [16]. To improve the thermal conductivity of microencapsulated PCM, nano-materials such as grapheme [17], graphene oxide [18] and diamond-nanoparticles [19] are introduced as additives in the shell. The other approach is the integration of paraffin in three dimensional (3D) porous materials, for example, 3D polymeric materials [20–22] and expanded graphite [23,24]. Expanded graphite is usually chosen as the supporting material in composite PCM, owing to its capacity of both encapsulation and heat transfer enhancement [25]. Although the above approaches can address the problems of leakage and poor thermal conductivity of paraffin, some defects are needed to be considered. The microencapsulated PCMs are in powder form and high brittleness [26], so that they are required form-shaping again in most applications. As for PCMs based on the expanded graphite, the form-shaped composite PCMs will become fragile as cyclic solid-liquid phase change [27]. Metal foam, as a kind of 3D rigid material, can greatly improve the mechanical performance and inhibit the fragility of the composite PCMs in the melting process [28]. Besides, the thermal conductivity of metal foam is an order of magnitude higher than that of PCMs [29]. Therefore, metal foam has the ability to improve the heat transfer rate of PCMs rapidly and stably, the thermal conductive pathways in the materials are not affected by liquid flow and volume expansion during the melting period of paraffin [30,31]. For example, Wang et al. [32] and Zheng et al. [33] studied the melting behavior of paraffin with and without metal foam and concluded that the use of metal foam could accelerate the melting process of paraffin and improve the temperature uniformity of composite PCM, suggesting the ability of metal foam to strengthen the thermal response of paraffin. However, the weakness of light-to-thermal energy conversion of paraffin has not been ameliorated.

To cover the shortage that PCMs hardly convert solar energy to thermal energy, researchers trend to choose the functional materials possessing the capacity of light-to-thermal energy conversion as the encapsulated materials [34–36]. Graphene aerogel (GA) exhibiting ultra-low density, strong capacity of adsorption and excellent ability of light-to-thermal energy conversion is considered as an optional encapsulated material [37]. Compositing PCMs with GA not only prevents the leakage of PCMs, but also makes the composite PCMs to convert light into thermal energy and store it in one step [38]. Mu and Li [39] synthesized composite PCMs with GA for solar energy conversion and storage. There was no liquid leakage from the composite PCMs detected during the heating process. And the maximum efficiency of light-thermal conversion obtained was 80.6%. This outcome suggested that incorporating GA could dramatically improve the light-to-heat conversion and prevent the leakage of PCMs. Tang et al. [40] prepared form-stable composite PCM using GA to adsorb paraffin for light-to-thermal energy conversion. Results showed the efficiency of light-to-thermal energy conversion was as high as 81.56%. The research demonstrated the more graphite sheets in the unit volume of the composite PCMs, absorption of light was enhanced and then light-to-thermal energy conversion was improved.

Comprehensively considering the advantages of metal foam and GA, this paper aims to develop a novel form-stable composite PCM with high thermal conductivity and capacity of light-to-thermal energy conversion for the application of solar energy. A porous matrix formed from GA loaded into the copper foam (CF) was prepared, and then paraffin was encapsulated in the GA/CF matrix by vacuum impregnation to fabricate paraffin/graphene aerogel/copper foam (paraffin/GA/CF) composite PCMs. Leakage test was carried out to verify the encapsulation stability

of the composite PCMs. Besides, the morphology and the thermal properties of the composite PCMs, including phase change temperature and enthalpy, thermal conductivity and thermal diffusion uniformity, were investigated. Finally, simulated solar illumination system was used to evaluate the potential application of the composite PCMs in light-to-thermal energy conversion and storage.

2. Materials and method

2.1. Materials

A series of paraffins with melting points of $\sim 60^\circ\text{C}$, $\sim 70^\circ\text{C}$, $\sim 80^\circ\text{C}$ and $\sim 90^\circ\text{C}$ were adopted here and labeled as PA60, PA70, PA80 and PA90, respectively. PA60 was provided by Aladdin Industrial Corporation (China). PA70, PA80 and PA90 were supplied by Shanghai Luer Chemical Trading Co., Ltd. Graphene oxide powders with a diameter of $0.5\text{--}5\text{ }\mu\text{m}$ and a thickness of $0.8\text{--}12\text{ nm}$ were purchased from Nanjing XFNANO Materials Tech Co., Ltd. L-ascorbic acid (L-AA) (purity $> 99.0\text{ wt}\%$) was from Sinopharm Chemical Reagent Co., Ltd respectively. Copper foam with dimension of $40\text{ mm} \times 40\text{ mm} \times 10\text{ mm}$, porosity of 0.98 and pore density of 40PPI was obtained from Suzhou JSD foam metal Co. Ltd.

2.2. Preparation of paraffin/GA/CF composite PCM

The preparation procedures of the paraffin/GA/CF composite PCMs is shown in Fig. 1. The reduction of the GO was performed via L-AA. In order to find the optimum adsorption capacity of paraffin, a series of GO aqueous solutions with different concentrations (10 mg/mL, 20 mg/mL, 30 mg/mL and 40 mg/mL) were prepared by ultrasound treatment for 1 h. L-AA at a mass ratio of 1:1 to GO was added to the GO aqueous solution under vigorous stirring. The GO was reduced at temperature of 20°C for 24 h. Removing 16 mL of graphene aqueous solution into square glass with internal dimension of $40\text{ mm} \times 40\text{ mm} \times 40\text{ mm}$, then copper foams were embedded in the graphene solutions, respectively. After freeze-drying for 48 h, GA/CF matrix was obtained. The characterizations of GO and GA can be found in the Supporting Information.

Paraffin was composited to the GA/CF matrix by vacuum impregnation method. Firstly, paraffin was heated with a temperature above its melting point to reach a liquid state. After that, the GA/CF matrix was immersed into the melted paraffin and maintain under vacuum condition for 1 h. Finally, the paraffin/GA/CF composite PCM was obtained after the sample was cooled to room temperature. All the composites were named in the format of PAX/GA (y)/CF, in which where PAX represents the type of paraffin and y was the value of GO concentration that was selected in the preparation stage.

To evaluate the encapsulation capability of GA for paraffin, a series of PA60/GA (y)/CF samples with different mass fractions of paraffin (Table 1) were prepared to perform leakage tests. An accelerated degradation method, which uses high temperature heating for a short period instead of low temperature heating for a long time [41], was employed for the tests. The original mass of the paraffin/GA/CF composite PCM and paraffin incorporated into the composite were represented by M and M_0 , respectively. The loss in the mass of paraffin after heating from 25°C to 80°C for 400 thermal cycles was denoted as M_l . The mass fractions of leaking paraffin (M_l/M_0) are plotted in Fig. 2. As it shows, the leakage percentages for paraffin in the selected cases increase with the number of thermal cycles. The values of the percentages will reach constant when sufficient thermal cycles are subjected to the composite samples due to the achievement of equilibrium state of mass transfer. Increasing the GO concentration from 10 to 30 mg/mL yields a significant drop in the leakage percentage of the paraffin in the composite, while a rise of the GO concentration from 30 to 40 mg/mL leads to the mass fraction of leaking paraffin increasing from 2.22 wt% to 2.56 wt%. The variation tendency of M_l/M_0 in response to the increased value of y implied an optimum GO concentration to provide suitable GA

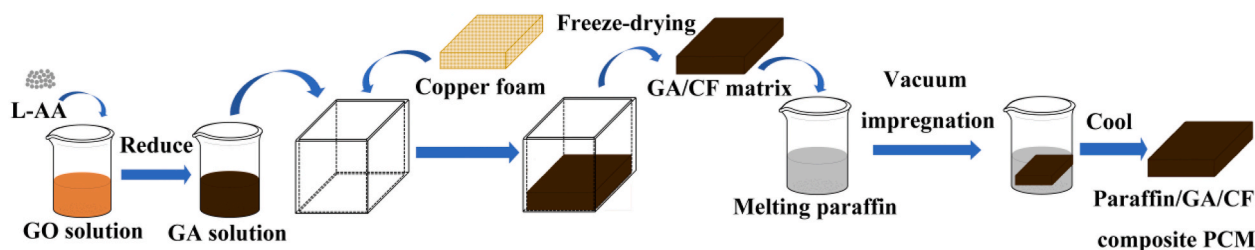


Fig. 1. Preparation procedures of paraffin/GA/CF composite PCMs.

Table 1

Paraffin/GA/CF composite PCMs prepared using different GO concentration.

Composites	GO concentration (mg/mL)	Mass fraction (wt%)		
		paraffin	GA	CF
PA60/GA (10)/CF	10	62.96	0.95	36.09
PA60/GA (20)/CF	20	66.60	1.64	31.76
PA60/GA (30)/CF	30	68.75	2.46	28.79
PA60/GA (40)/CF	40	67.58	4.27	28.15

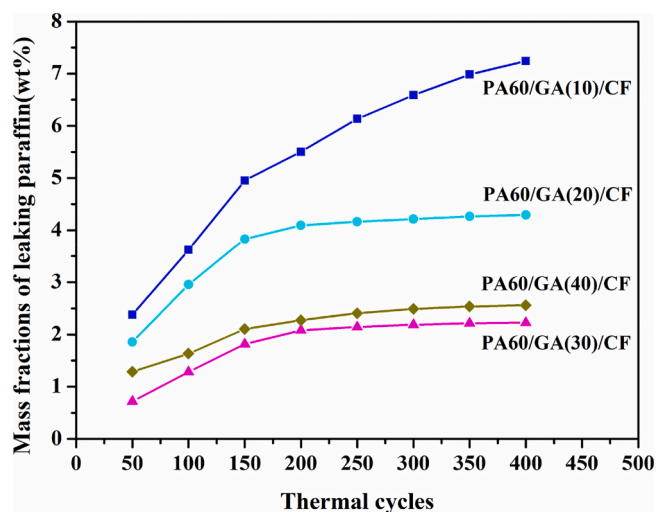


Fig. 2. Leakage performance of the PA60/GA (y)/CF composite PCMs.

structures for paraffin encapsulation. The concentration of 30 mg/mL for GO, therefore, was selected to prepare paraffin/GA/CF composite PCM.

In order to investigate the effects of phase change temperatures of the paraffin/GA/CF composite PCMs on the light-thermal energy conversion and storage performances of them, composite samples containing 65 wt% paraffin with different melting points were prepared. The obtained samples were labeled as PA60/GA (30)/CF, PA70/GA (30)/CF, PA80/GA (30)/CF and PA90/GA (30)/CF, respectively. The detailed compositions of the samples are listed in Table 2.

Table 2

Composite PCMs prepared using paraffins with various melting points.

Samples	Mass fraction (wt%)		
	paraffin	GA	CF
PA60/GA (30)/CF	65.47	5.19	29.34
PA70/GA (30)/CF	65.61	5.26	29.12
PA80/GA (30)/CF	65.09	4.30	30.61
PA90/GA (30)/CF	65.96	4.57	29.47

2.3. Instruments and measurements

Morphology observations of the paraffin/GA/CF composite PCMs were performed using a scanning electron microscopy (SEM, SU8220, Hitachi High-Technologies Co., Ltd, Japan) the composite samples were coated with thin gold layers to improve conductivity and prevent charging. Fourier transform infrared spectroscopy (FTIR, VERTEX 70, Bruker Co. Ltd., Germany) and X-ray diffractometer (X'Pert3 Powder, PANalytical B.V., Holland) were employed to characterize the structure of paraffin, GA/CF matrix and paraffin/GA/CF composite PCM. To analyze the phase change temperatures of the as-prepared material samples, a differential scanning calorimetry (DSC 214 Polyma, NETZSCH Instruments, Germany) was used. The scanning temperature range for each DSC test was from 0 to 120 °C and the heating and cooling rate for each sample was set to be 1 °C/min under nitrogen atmosphere. Thermal conductivity was determined by using Hot disk thermal analyzer (TPS2500, Hot Disk Inc., Sweden) based on the transient plane source (TPS) method. The precision of the measurements is within $\pm 2\%$. Infrared thermal imager (Ti9, FLUKE Inc., America) was employed to monitor the temperature distributions on the surface of the composite PCM sample. To evaluate the light-thermal conversion efficiency of the composite PCM, paraffin and the composite PCM were directly illuminated by a simulated solar light system (CEL-HXUV300, CEAUKIGHT, China). A data acquisition unit (Agilent 34970A), equipped with K-type thermocouples was used to collect temperature evolution of paraffin and the composite PCM. The accuracy of the thermocouples was within ± 0.1 °C.

3. Results and discussions

3.1. Morphology observation

The SEM images of GA/CF matrix and paraffin/GA/CF composite PCM (PA60/GA (30)/CF) are shown in Fig. 3. From Fig. 3 (a) and Fig. 3 (b), it can be seen that, in the structures of GA/CF matrix, the macropores of the copper foam were successfully impregnated with porous GA which can provide pores with micro-sizes for paraffin absorption and encapsulation. The margins of the copper framework within the GA regions seem to be very clear and sharp, indicating the GA infiltration process did not destroy the copper foam skeleton. The magnified image of the GA/CF matrix (Fig. 3(c)) offers a better view of the micro-pores in GA. Stacked graphene layers with wrinkled surfaces that help to form the pore walls of GA are observed clearly. Fig. 3 (d)–(f) displays the morphology of the paraffin/GA/CF composite PCM. As we can see, most of the pores in the GA/CF matrix disappear due to the impregnation of paraffin. The existence of paraffin in the composite makes the wrinkled surfaces and folded edges of the graphene layers of GA very smooth. A part of the GA pores with large sizes are still exposed with little amount of paraffin inside. Possible reason could be the large pores of GA provided insufficient capillary forces to confine the liquid paraffin. Above results indicated that the pores of GA played a dominant role in prevention of leakage problem of the paraffin/GA/CF composite PCM.

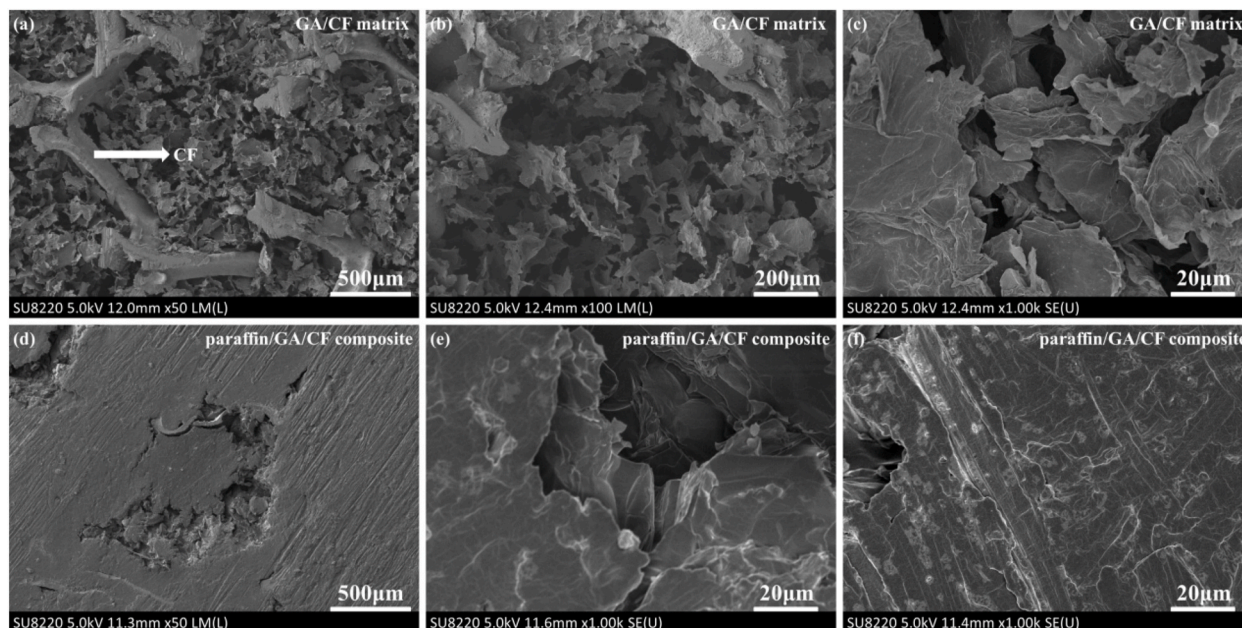


Fig. 3. SEM images of (a, b, c) GA/CF matrix and (d, e and f) the paraffin/GA/CF composite PCM.

3.2. Structure characterization

FTIR spectroscopy was used to analyze the chemical structure of paraffin/GA/CF composite PCM. Fig. 4 (a) displays the FTIR spectra of GA/CF matrix, paraffin and the composite PCM (PA60/GA (30)/CF). For paraffin, there are five distinct characteristic peaks. The peaks at 2918 cm^{-1} and 2849 cm^{-1} can be assigned to the stretching vibrations of C–H in $-\text{CH}_3$ and $-\text{CH}_2-$ groups, respectively. The peaks at 1463 cm^{-1} and 1378 cm^{-1} are resulted from the C–H bending vibrations of $-\text{CH}_2-$ and $-\text{CH}_3$. The 719 cm^{-1} peak is the in-plane rocking vibrations of C–H. All these peaks have confirmed the existence of long alkyl chains and agree with the structures of paraffin [42]. As for GA/CF matrix, stretching vibration peaks of C–H are observed. The peak representing the stretching vibration of C=C is found at 1530 cm^{-1} . Considering the intensities of these peaks are low, the amounts of $-\text{CH}_2-$, $-\text{CH}_3$ and C=C groups in the GA/CF matrix could be very small [43,44]. When it comes to the composite sample, it is seen that all the characteristic peaks of paraffin and GA/CF matrix can be found in the spectrum of the paraffin/GA/CF composite PCM. No new peak has been generated when paraffin is composited with GA/CF matrix. These results imply that, in the composite PCM, the paraffin was only physically blended with GA/CF matrix.

XRD tests were employed to analyze the crystalline features of paraffin in the paraffin/GA/CF composite PCM. It is seen in Fig. 4 (b) that there are two sharp XRD characteristic diffraction peaks for paraffin which can be found at $2\theta = 21.3^\circ$ and $2\theta = 23.7^\circ$ respectively. The intensities of the two peaks are very high, indicating the formation of crystals in the bulk sample of paraffin [42]. Three sharp diffraction peaks are observed at $2\theta = 43.2^\circ$, 50.4° and 70.0° in the XRD image of GA/CF matrix. Considering there is only one small broad peak in the XRD pattern of GA (Fig. S2), the three sharp peaks might come from the crystals in the copper foam (Fig. S3). As for XRD result of the paraffin/GA/CF composite PCM, all the characteristic diffraction peaks of both paraffin and GA/CF matrix can be found. No new diffraction peaks is detected and no shifts of the characteristic peaks of paraffin and GA/CF matrix can be observed in the XRD spectra of the composite. These results combined with the FTIR data proved that there was no chemical interaction between paraffin and GA/CF matrix in the paraffin/GA/CF composite PCM.

3.3. Phase change property analysis

Several types of paraffin with different phase change temperatures were employed to fabricate the paraffin/GA/CF composite PCM. The

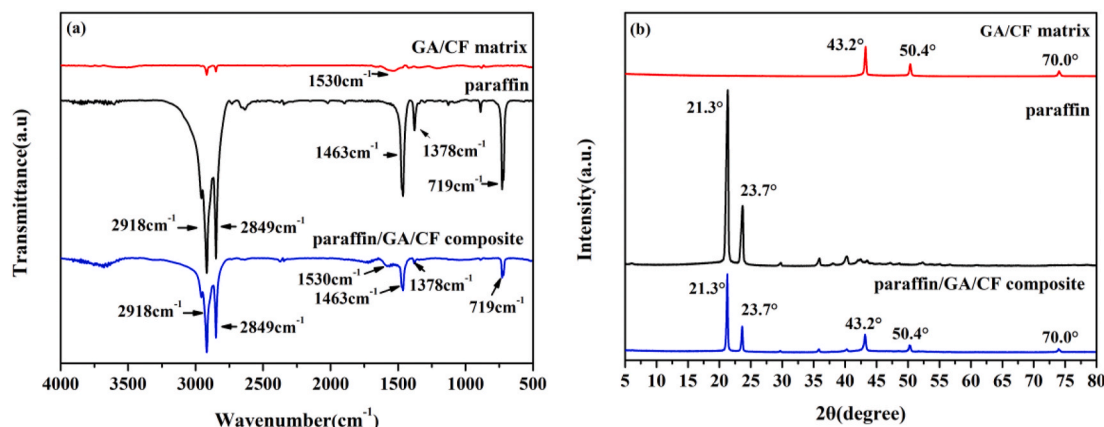


Fig. 4. (a) FTIR spectra and (b) XRD patterns of GA/CF matrix, paraffin and paraffin/GA/CF composite PCM.

phase change properties of the selected paraffin including phase change temperature and latent heat were determined through DSC tests. The resulting DSC curves are shown in Fig. 5. It is observed that the melting and crystallization peaks of each type of paraffin are located at closer positions, indicating that paraffin is free from super-cooling problem. The melting and crystallization temperatures of PA60, PA70, PA80 and PA90 are marked in the figure and positioned at around 60 °C, 70 °C, 80 °C and 90 °C, respectively, which right meets our expectations. Multiple peaks are observed for some of types of paraffin. According to previous studies [45,46], the phase change of some pure long chain alkane (such as n-octacosane) may give rise to several types rotating phases. The multiple phase change peaks of the paraffin could be resulted from the phase transitions among the crystal phase, rotating phases and the liquid phase of paraffin. The phase change latent heat of these types of paraffin derived from the presented DSC curves are summarized in Table 3. Based on our previous discussions, it is noted that in the paraffin/GA/CF composite PCM, paraffin which serves as the thermal energy storage media only physically blended with the GA/CF matrix. The GA/CF matrix plays the roles of providing heat transfer pathways and shape-stabilized supporting skeleton both in micro- and macro-scales for paraffin. Since numerous studies indicated that the latent heat of most composite PCM developed through physical blending approaches is positively related to the mass fraction and latent heat of the thermal energy storage media [24], the predicted melting and freezing latent heat of the as-prepared several types of paraffin/GA/CF composite PCMs (ΔH_m and ΔH_c) can be obtained through following the same calculation rule. The results can be found in Table 3.

3.4. Heat conductive behavior analysis

Thermal conductivities of the pristine paraffins and the as-prepared series of bulk paraffin/GA/CF composite PCMs are measured at room temperature (about 25 °C) and the results are given in Fig. 6. It is seen

Table 3

DSC data of paraffin and composite PCMs with different melting temperature.

Samples	ΔH_m (J/g)	ΔH_c (J/g)	Samples	ΔH_m (J/g)	ΔH_c (J/g)
PA60	204.1	203.7	PA60/GA (30)/CF	133.62	133.36
PA70	266.1	263.8	PA70/GA (30)/CF	174.59	173.08
PA80	221.3	221.0	PA80/GA (30)/CF	144.04	143.85
PA90	213.3	207.5	PA90/GA (30)/CF	140.69	136.87

that, in comparison to pristine paraffins, all types of paraffin/GA/CF composite PCMs possess much higher thermal conductivity (more than 9 times higher). As GA and CF exhibit much greater heat conductive ability than paraffin [29,47], the graphene layers and copper wires comes from the GA and CF would form continuous pathways for heat transfer in the composite PCM bulk samples, consequently leading to significant enhancement in the thermal performance of the paraffin in the composite. Besides, for the different types of paraffins, the thermal conductivities of them are around 0.3 W/(m·K). A very small difference in thermal conductivity among these paraffins are observed. The thermal conductivities of the corresponding paraffin/GA/CF composite PCMs are also very close (around 3.0 W/(m·K)). These indicate that the thermal conductivity enhancement effects brought about by the GA/CF matrix to different types of paraffins are similar.

Fig. 7 shows the infrared thermal images of PA60/GA (30)/CF as a representative to demonstrate whether the composite PCM with a thermal conductivity of around 3.0 W/(m·K) could achieve uniform temperature distributions during the energy storage process. The thermal energy absorbed by the composite PCM was generated by an electronic heater with a constant temperature of 80 °C (Fig. 7(a)). Infrared

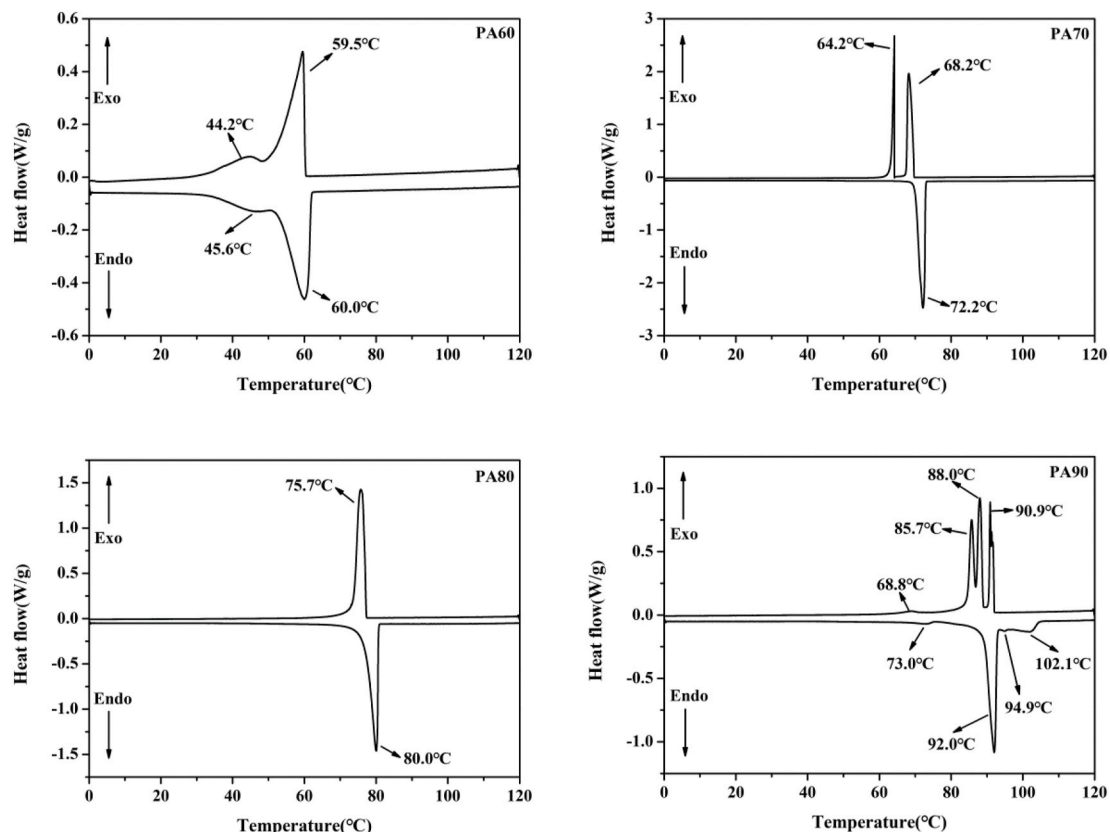


Fig. 5. DSC curves of different types of paraffin.

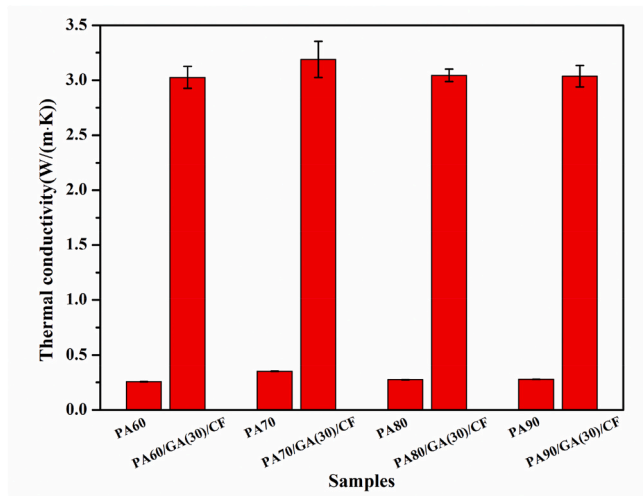


Fig. 6. Thermal conductivity of paraffin and paraffin/GA/CF composite PCMs.

thermal imager was recorded every 5min. It is seen in Fig. 7 (b), as the heat storage process of the composite PCM progresses, the overall temperature of the composite bulk temperature increases. At $t = 5$ min, 10 min and 15 min, the patterns with higher temperature marked denote the shapes of the GA/CF matrix and indicate that the heat transfer in the skeleton of the GA/CF matrix is faster than that in the paraffin. The regions with lower temperature marked in the composite block are paraffin and demonstrate about 5°C lower than the temperature of the GA/CF matrix. With the heat being transferred from the bottom surface of the bulk composite sample to the upper surface, the infrared thermal patterns of both the bottom and upper sides of the block sample are very close. This means relatively homogeneous heat transfer and heat absorption were achieved inside the block due to the heat transfer pathways within the GA/CF matrix that significantly facilitate the simultaneously heating of the sample. At $t = 20$ min, the composite reaches a thermal equilibrium state and both the paraffin and matrix within the composite maintain at 80°C . Above results indicated that with the assistance of the GA/CF matrix, a relatively uniform temperature distribution within the paraffin/GA/CF composite PCM during thermal energy storage process can be found.

3.5. Light-to-thermal energy conversion performance evaluation

For the purpose of evaluating the light-to-thermal energy conversion

performance of the paraffin/GA/CF composite PCM, PA60, PA60/GA (30)/CF, PA70/GA (30)/CF, PA80/GA (30)/CF and PA90/GA (30)/CF were tested under simulated solar irradiation (AM1.5) at a constant intensity of 100 mW/cm^2 (Fig. 8). The temperature variations of paraffin and the composite PCMs are recorded in Fig. 9. It can be seen that, the increase of the irradiation time, the surface temperatures of paraffin and the composite PCM bulk samples increase accordingly. The temperature increasing rate of the composite PCMs is obviously higher than that of paraffin. Take PA60 and PA60/GA (30)/CF as examples, 150 min to can allow PA60/GA (30)/CF to increase its temperature from 30°C to 104°C but only can enable PA60 to reach 61°C . Due to the graphene sheets in GA are effective photon captors and molecular heaters and can absorb and convert solar energy into thermal energy, the paraffin/GA/CF composite PCM would receive more heat supply for energy storage. As for PA60, the only heat source is the infrared radiation from the sunlight which will lead to a low heating power density for PA60 sample. A lower temperature is observed for this material consequently. In addition, for the composite PCMs, there are obvious pseudo-platforms in their temperature profiles, which corresponds to the phase transition of paraffins. The thermal energy obtained from the conversion effect of GA and directly from the solar radiation could be stored in a form of latent heat during this phase transition process. When the light source is turned off, the temperature of the composite PCMs drops sharply, and then the phase change pseudo-platform appears. The pseudo-platform in the cooling process shows the temperatures of the composite PCMs maintain at a relatively high level in a wide range of cooling time, indicating the good heat preservation ability of the composite PCM. Besides, it should be noted that phase transition pseudo-platform also appears in the thermal profiles of PA60 during illumination process. Because the low heating power and the poor heat transfer property of paraffin, the latent thermal energy process of PA60 lasts for more than 110 min. A poor solar energy storage performance of paraffin as well as the requirements of modifications of this material using GA/CF matrix was highlighted.

The light-to-thermal energy conversion efficiency (η), which was used to compare the light-to-thermal energy conversion ability of the composite PCMs, can be calculated by the following equation [38]:

$$\eta = \frac{m\Delta H}{\rho S(t_f - t_s)} \quad (1)$$

where m is the mass of the sample, ΔH is the latent heat, ρ is the intensity of light irradiation of the simulated solar light source and the value of ρ is set to be 100 mW/cm^2 , in this work, S is the surface area of sample, t_f and t_s are the phase transition time before and after the phase change, respectively. The calculated light-to-thermal energy conversion

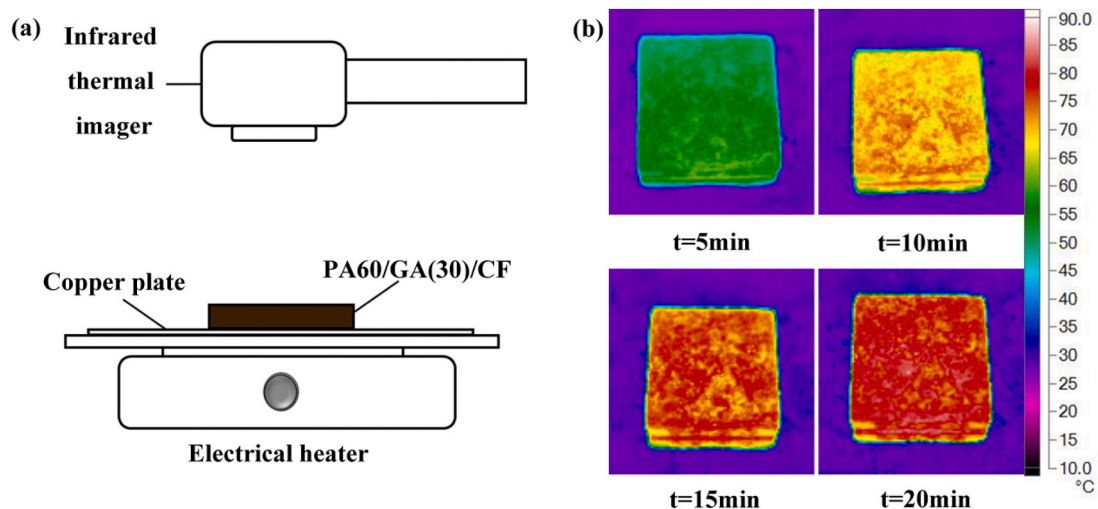


Fig. 7. (A) Schematic illustration of infrared thermal imaging and (b) infrared thermal images of paraffin/GA/CF composite PCM.

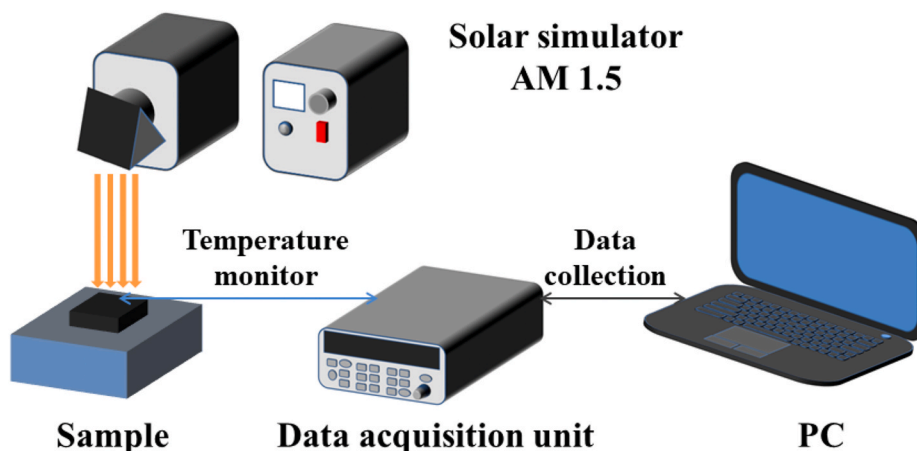


Fig. 8. Experimental setup of light-to-thermal energy conversion of PCMs.

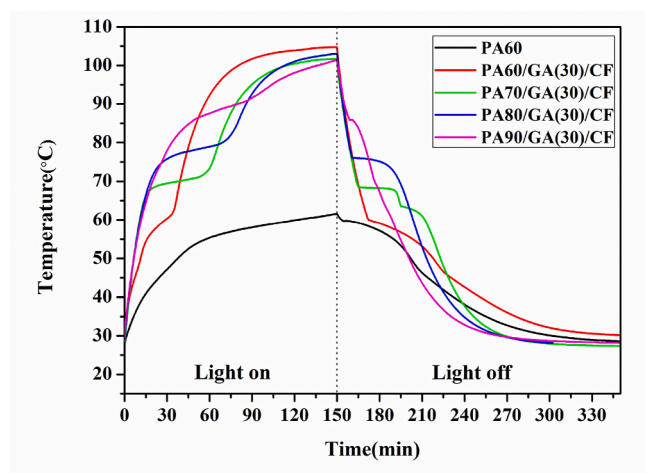


Fig. 9. Temperature vs. time curves of paraffin and paraffin/GA/CF composite PCM.

efficiencies of PA60/GA (30)/CF, PA70/GA (30)/CF, PA80/GA (30)/CF and PA90/GA (30)/CF are 97%, 81%, 71% and 66%, respectively. Obviously, PA60/GA (30)/CF composite has the highest η value. It is because the latent heat storage process of the paraffin/GA/CF composite PCM occurs at around 30 °C which can maintain a small temperature difference between the PCM block and ambient. The smallest amount of heat for dissipation to the ambient therefore could be obtained by PA60/GA (30)/CF in comparison to other composite samples.

4. Conclusion

In this work, a novel form-stable paraffin/GA/CF composite PCM with good heat conductive property and high efficiency of light-to-thermal energy conversion was developed for solar energy storage. SEM observations of the internal microstructures of the composite PCM demonstrated that paraffin impregnated into the micro-pores within the GA/CF matrix. Leakage proof property was found for the composite PCM due to the encapsulation function of the micro-pores formed by the graphene sheets and copper skeleton. Benefiting from the continuous conductive interconnected netlike structure of CF, continuous heat transfer pathways are built within the composite and the thermal conductivity of the composite PCM is significantly improved. Relatively uniform temperature distributions are found for the composite sample because of the improved thermal conductivity. Moreover, high light-to-thermal energy conversion efficiency and the latent heat suggest that the

composite PCMs can efficiently convert a large amount of solar energy into thermal energy and store it at the same time, meaning the increase of utilization efficiency of solar energy. Above results implied that the paraffin/GA/CF composite PCMs have great potential for application on solar energy conversion and storage.

CRediT authorship contribution statement

Xiangjun Zheng: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Xuenong Gao:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Zhaowen Huang:** Methodology, Writing – review & editing, Visualization. **Zhongping Li:** Formal analysis. **Yutang Fang:** Supervision. **Zhengguo Zhang:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The author, Zhengguo Zhang, is the editor of the *Solar Energy Materials and Solar Cells*.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.solmat.2021.111083>.

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